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Inventor(s)	Griffin; Leigh et al.

Dynamically adjusting biases on quantum bits based on detected events

Abstract

Biases on quantum bits can be dynamically adjusted based on events. For example, a system can detect an event related to a service executing in a computing environment. The service can rely on a state of a quantum bit of a quantum computing subsystem in performing a computing operation. The system can determine an amount of bias to apply to the quantum bit based on the event. The amount of bias can be configured to modify a result of the computing operation. The system can transmit a control signal configured to cause the quantum computing subsystem to apply the amount of bias to the quantum bit (e.g., while the service is executing).

Inventors:	Griffin; Leigh (Waterford, IE), Coady; Stephen (Waterford, IE)
Applicant:	RED HAT, INC. (Raleigh, NC)
Family ID:	83080734
Assignee:	Red Hat, Inc. (Raleigh, NC)
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Primary Examiner: Memula; Suresh

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to quantum computing. More specifically, but not by

way of limitation, this disclosure relates to dynamically adjusting biases on quantum bits based on detected events.

BACKGROUND

(2) Quantum computing subsystems harness quantum mechanics to provide significant advances in computation to solve problems. The main building block of a quantum computing subsystem is a quantum bit (or “qubit”). Quantum bits serve as the basic unit of information in quantum computing subsystems, much like how binary bits serve as the basic unit of information in classical computers. Common types of quantum bits include charge qubits and flux qubits.

(3) A single quantum bit can have two or more discrete energy states, which are often referred to as basis states. The state of a quantum bit at a given instant in time can be any superposition of two basis states, which means that a quantum bit can be in the two basis states at the same time. This is fundamentally different from how a conventional binary bit operates on a classical computer, whereby the bit can only be in a single state (a 0 state or a 1 state) at a given instant in time.

(4) Quantum bits are often used to perform quantum computations. To complete a quantum computation using a quantum bit, the state of the quantum bit is typically measured (e.g., read out). Due to certain physical phenomena, the quantum nature of a quantum bit may be temporarily lost during the measurement process, causing the superposition of the two basis states to collapse into either one basis state or the other.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of an example of a system for dynamically adjusting biases on quantum bits based on detected events according to some aspects of the present disclosure.

(2) FIG. 2 is a block diagram of an example of a predefined mapping according to some aspects of the present disclosure.

(3) FIG. 3 is a block diagram of an example of a system for implementing some aspects of the present disclosure.

(4) FIG. 4 is a flow chart of an example of a process for dynamically adjusting biases on quantum bits in response to detecting events according to some aspects of the present disclosure.

(5) FIG. 5 is a block diagram of an example of a system for controlling access to a resource by dynamically adjusting biases on quantum bits according to some aspects of the present disclosure.

DETAILED DESCRIPTION

(6) A quantum computing system can include quantum bits that may serve as the basic unit of information within the quantum computing system. The states (e.g., basis states) of the quantum bits may be set in specific ways to configure the quantum computing system. Conventionally, the states of the quantum bits are preset and remain relatively fixed during the lifetime of a software service executed by the quantum computing system. For example, a quantum computing system may execute a software service to perform operations. Examples of the software service can include a microservice, serverless function, or application. Prior to executing the software service, the states of the quantum bits may be set in a particular way to configure the quantum computing system. And the states of the quantum bits may remain relatively fixed while the software service executes. The reason that the states of the quantum bits are typically kept constant during the software service's runtime is so that computing operations performed by the software service yield consistent results. Modifying the states during runtime can lead to inconsistent results, for example if the same computing operation is executed twice based on the same inputs. But there are some circumstances where modifying the quantum bit's states during runtime may be desirable, for example to guide a computing operation's result toward or away from a target value. And convention quantum computing systems lack the flexibility to do so.

(7) Some examples of the present disclosure can overcome one or more of the abovementioned problems by dynamically adjusting biases on quantum bits in response to detecting events related to a software service, so as to influence a result of a computing operation implemented by the software service at runtime. More specifically, a computer can include a software service that is configured to execute a computing operation. A result of the computing operation can depend on a state of a quantum bit of a quantum computing subsystem. During the runtime of the software service, the computer can detect one or more events that relate to the computing operation. In response to detecting such events, the computer can interact with the quantum computing subsystem to adjust a bias on the quantum bit in a way that modifies a future result of the computing operation. Different events may lead to different adjustments to the bias, thereby causing the computing operation to produce different results in response to different events. In this way, the computer can dynamically adjust the bias on the quantum bit during the runtime of the software service to influence the results of the computing operation.

(8) In some examples, the computer can determine how to adjust the bias on the quantum bit based on a predefined mapping. The predefined mapping may include relationships between preselected events and bias amounts or adjustments. The computer can access the predefined mapping to determine an amount of bias that is correlated to a detected event. The computer can then interact with the quantum computing subsystem to adjust the bias on the quantum bit based on (e.g., to match) the amount of bias defined in the mapping. Because the computing operation of the software service can depend on the state of the quantum bit, adjusting the bias on the quantum bit may affect a subsequent result of the computing operation.

(9) In other examples, the computer can determine how to adjust the bias on the quantum bit based on a model. The model may be a machine-learning model, such as a neural network, classifier, or support vector machine. The model can be configured to receive event data as input and generate an output indicating a bias amount or adjustment. The computer can supply event data associated with a detected event as input to the model and receive as output from the model an amount of bias to apply to a quantum bit. The computer can then interact with the quantum computing subsystem to adjust the bias on the quantum bit based on the amount of bias indicated by the model. Because the computing operation of the software service can depend on the state of the quantum bit, adjusting the bias on the quantum bit may affect a subsequent result of the computing operation.

(10) These illustrative examples are given to introduce the reader to the general subject matter discussed here and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements but, like the illustrative examples, should not be used to limit the present disclosure.

(11) FIG. 1 is a block diagram of an example of a system **100** for dynamically adjusting biases on quantum bits in response to detecting events according to some aspects of the present disclosure. The system **100** can include a computer **102**, such as a laptop computer, desktop computer, or server. The computer **102** can include a computing environment **104** for running one or more software services, such as software service **108**. Examples of the software service **108** can include a gaming application, a navigation application, a word-processing application, a data-processing application, etc. The software service **108** may be any suitable type of software program, such as a microservice, a serverless function, or an application.

(12) The computer **102** can also include various hardware. For example, the computer **102** can include a processor **112** communicatively coupled to a memory **126**. The processor **112** can include one processing device or multiple processing devices. Non-limiting examples of the processor **112** include a Field-Programmable Gate Array (FPGA), an application-specific integrated circuit (ASIC), a microprocessor, etc. The processor **112** can execute instructions stored in the memory **126** to perform the operations. In some examples, the instructions can include processor-specific instructions generated by a compiler or an interpreter from code written in any suitable computer-

programming language, such as C, C++, C #, etc.

(13) Memory **126** can include one memory device or multiple memory devices. The memory **126** can be non-volatile and may include any type of memory device that retains stored information when powered off. Non-limiting examples of the memory **126** include electrically erasable and programmable read-only memory (EEPROM), flash memory, or any other type of non-volatile memory. At least some of the memory **126** can include a non-transitory computer-readable medium from which the processor **112** can read instructions. A computer-readable medium can include electronic, optical, magnetic, or other storage devices capable of providing the processor **112** with computer-readable instructions or other program code. Examples of a computer-readable medium can include magnetic disks, memory chips, ROM, random-access memory RAM, an ASIC, a configured processor, optical storage, or any other medium from which a computer processor can read instructions.

(14) The processor **112** can also be communicatively coupled to a quantum interface **114** for interfacing with a quantum computing subsystem **116**. The quantum interface **114** can include hardware, software, firmware, or any combination thereof for interacting with the quantum computing subsystem **116**. The quantum interface **114** may receive control signals **130** from the processor **112** and operate the quantum computing subsystem **116** based on the control signals **130**.

(15) The quantum computing subsystem **116** can include a group **118** of quantum bits **120a-n**. The group **118** can include any number and combination of quantum bits **120a-n**. In some cases, the group **118** can include thousands of quantum bits **120a-n**. The quantum bits **120a-n** may be superconducting quantum bits, in some examples.

(16) The quantum computing subsystem **116** can also include a biasing circuit **122** associated with the quantum bits **120a-n**. The biasing circuit **122** is hardware for applying one or more biases to one or more of the quantum bits **120a-n**. The biases can influence the states of the quantum bits **120a-n**. Examples of the biases can include a magnetic field or an electric charge configured to influence the state of a corresponding quantum bit.

(17) To generate the biases, the biasing circuit **122** can include one or more biasing devices **124a-n**. The biasing devices **124a-n** are hardware configured to apply the biases to the quantum bits **120a-n**. The biasing devices **124a-n** may include electrical circuit components for generating the biases. In the example shown in FIG. 1, each of the quantum bits **120a-n** has a corresponding biasing device **124a-n** that is configured to apply a bias to that quantum bit. The bias is depicted in FIG. 1 as a double-headed arrow between the biasing device and the quantum bit. But in other examples, the biasing circuit **122** may be configured in other ways. For example, multiple quantum bits **120a-b** may share a single biasing device **124a**, where that biasing device **124a** can apply one or more biases to the multiple quantum bits **120a-b** sequentially or concurrently.

(18) The biasing devices **124a-n** can be made of any suitable hardware components that may be programmatically controlled. For example, a biasing device **124a** can include a loop of metal in proximity to a quantum bit **120a** to generate a magnetic field on the quantum bit **120a**. The loop of metal may include a Josephson junction. The characteristics of the magnetic field can depend on the properties (e.g., magnitude, frequency, and phase) of the current supplied to the loop, where the current properties can be programmatically controlled. The current properties may be programmatically controlled by the processor **112** of the computer **102** or by any other suitable hardware, such as a control system that is internal to the quantum computing subsystem **116**. In this way, each of the biasing devices **124a-n** can be individually and programmatically controlled to apply a certain amount of bias on at least one corresponding quantum bit. Applying the bias to the quantum bit can influence the state of the quantum bit, for example by causing the quantum bit to have a higher probability of being in a target basis state or landing in the target basis state subsequent to a readout operation.

(19) In some examples, the computer **102** may include an event detector **106**. The event detector **106** may be software configured for detecting events that are associated with the software service

108. The event detector **106** can detect the events based on information from one or more data sources **134**, such as websites, RSS feeds, file servers, video or audio streams, sensors, or any combination of these. The event detector **106** may receive the information from the data sources **134** via one or more networks **132**, such as a local area network or the Internet. The event detector **106** can then analyze (e.g., parse) the information to detect the events.

(20) The events can include any type of events occurring internally or externally to the service **108** that may impact the operation of the software service **108**. In some examples, the events may originate from the software service **108**. For instance, the events can include program events triggered by the software service **108**. Examples of such program events can include errors, the execution or completion of certain program functions, or the generation of certain outputs. In other examples, the events may originate outside the software service **108** and may occur independently of the software service **108**. For instance, the events may be virtual events occurring elsewhere within the computing environment **104** or elsewhere in digital space (e.g., within another computing environment that is external to the computer **102**). Examples of such virtual events may include cybersecurity events, stock market events, video game events, and social media events. In some examples, the events may include physical events occurring in the real space (the real world), such as news events or sensed events. Examples of news events can include sporting events, vehicle accidents, weather events, and corporate events. Examples of sensed events can be events detected based on sensor signals from one or more sensors, such as thermometers, gyroscopes, accelerometers, inclinometers, cameras, microphones, or any combination of these.

(21) In some examples, the processor **112** can execute the event detector **106** to detect an event **138**. Although the event **138** is depicted in FIG. 1 as being internal to the event detector **106**, this is merely meant to symbolize that the event detector **106** detected the event **138**, rather than that the event **138** originated from the event detector **106**. The event detector **106** can detect the event **138** while the software service **108** is running. In some examples, the event **138** may be related to a computing operation **110** that is currently being performed by the software service **108**. Alternatively, the event **138** may be related to a computing operation **110** that will be performed by the software service **108** in the future.

(22) The computing operation **110** may be any suitable computing operation that relies on one or more of the quantum bits **120a-n** in generating a result. In particular, the computing operation **110** may be configured to generate a certain result based on the basis states of the quantum bits **120a-n**. For example, the software service **108** can be a navigation service and the computing operation **110** may be a navigation operation for determining a driving route from a starting location to a destination location. Part of the process for determining the driving route may involve a computation that relies upon one or more basis states of one or more quantum bits **120a-n**. As another example, the software service **108** can be an access control service and the computing operation **110** may be an access control operation for controlling access to a resource, such as a virtual resource or a physical resource. Examples of the virtual resource may be a secure file or a secure server. Examples of the physical resource may be a secure object or a secure physical location. Part of the process for determining whether access is to be granted to the resource may involve a computation that relies upon one or more basis states of one or more quantum bits **120a-n**.

(23) In response to detecting the event **138**, the processor **112** can determine how to adjust one or more biases on one or more of the quantum bits **120a-n** that are relied upon by the computing operation **110**. Adjusting the biases can influence the computing operation **110** in such a way that the computing operation **110** is more likely to achieve a target result. The specific quantum bits that are to be impacted, and the amount of bias to be applied to those quantum bits, can be determined based on the detected event **138**. For example, the processor **112** can access a mapping **128**, which may be predefined, to determine bias amounts to apply in response to detecting events.

(24) In some examples, the mapping **128** can be service-specific or domain-specific. For example,

the computer **102** may have access to multiple mappings, where each mapping is designed to be used with a corresponding software service. The mapping **128** may be specifically designed for use with the software service **108**, and other software services may have their own respective mappings. Such mappings may be created by the developers of the software services or other developers more familiar with quantum mechanics.

(25) One example of the mapping **128** is shown in FIG. **2**. As shown, the mapping **128** can include relationships between events, bias amounts or adjustments **204a-n**, and quantum bits. The events can be represented in the mapping **128** using event identifiers **202a-n**. The quantum bits can be represented in the mapping **128** using quantum bit identifiers **206a-n**. As shown, there can be one-to-one or one-to-many relationships between events, bias amounts, quantum bits, or any combination of these. Based on the mapping **128**, the processor **112** can determine which bias adjustments are to be made to which quantum bits based on the detected event **138**. For example, the detected event **138** may correspond to event identifier **202a**. So, the processor **112** can use a first correlation in the mapping **128** between event identifier **202a** and bias amount **204a** to determine that the bias amount **204a** is to be applied to one or more quantum bits in response to detecting the event **138**. The processor **112** can also use a second correlation in the mapping **128** between the bias amount **204a** and the quantum bit identifier **206a** to determine that the bias amount **204a** is to be applied to the particular quantum bit **120a** associated with the quantum bit identifier **206a**.

(26) Referring back to FIG. **1**, the processor **112** can additionally or alternatively determine the bias amounts to apply and the corresponding quantum bits in other ways. For example, the processor **112** can execute a model **136**, such as a machine-learning model. Examples of a machine-learning model can include a neural network, a classifier, or a support vector machine. The model **136** can be configured to receive inputs and generate an output indicating one or more bias amounts and one or more quantum bits to which the one or more bias amounts are to be applied. In some examples, the inputs can include event data characterizing the detected event **138**. The output from the model **136** can indicate bias amounts and corresponding quantum bits based on the event data.

(27) The model **136** may be trained prior to usage. A supervised or unsupervised training process may be used to train the model **136**. The model **136** may be trained using any suitable training data. In some examples, the training data may include relationships between the states of one or more quantum bits **120a-b**, bias amounts applied to the one or more quantum bits by the biasing devices **125a-n**, and outputs of the computing operation **110**. For instance, the training data may include thousands or millions of entries, where each entry includes one or more states of one or more quantum bits **120a-b** at a given instant in time, one or more bias amounts applied to the one or more quantum bits at that instant in time, and an output of the computing operation **110**. From the training data, the model **136** may learn how different biases impact the states of the quantum bits and how the states of the quantum bits impact the output of the computing operation **110**. Once trained, the model **136** may be able to estimate how to adjust the biases applied to certain quantum bits to yield a desired result from the computing operation **110**. The model **136** may provide this information as output, which the processor **112** can use to configure the biases on the quantum bits **120a-n** in a way that is more likely to achieve the desired result.

(28) Having determined the bias amounts to apply and the corresponding quantum bits, the processor **112** can next generate one or more control signals **130** based on the determined bias amounts and the determined quantum bits. The control signals **130** can be configured to cause the quantum computing subsystem **116** to apply the determined bias amounts to the determined quantum bits. For example, the processor **112** can transmit the control signals **130** to the quantum interface **114**, which can receive the control signals **130** and responsively operate the biasing circuit **122** such that the biasing circuit **122** applies the determined bias amounts to the determined quantum bits. This may involve the quantum interface **114** generating and transmitting secondary control signals **140** to the quantum computing subsystem **116** based on the control signals **130** from

the processor **112**. The secondary control signals **140** may cause the biasing circuit **122** (e.g., one or more biasing devices therein) to apply the determined bias amounts to the determined quantum bits.

(29) The processor **112** can transmit the control signals **130** while the software service **108** is running to provide runtime control over the service **108**. This may cause the computing operation **110** to produce a different result than may otherwise be produced absent the control signals **130**, because the computing operation's result is dependent on the states of the quantum bits to which the bias amounts are applied. In this way, the results of the computing operation's results may be dynamically influenced in response to events detected by the event detector **106**. Thus, the same inputs to the computing operation **110** may yield different results depending on the events occurring prior to each iteration of the computing operation **110**.

(30) As noted above, applying the biases to the quantum bits can influence the results of the computing operation **110** in a various way, for example such that the computing operation **110** is more likely generate a target result. One such example may occur within the context of a navigation service. In that context, the processor **112** may detect an event **138** (e.g., via the event detector **106**) involving a car accident on a highway. In response to detecting the event **138**, the processor **112** may determine an amount of bias to apply to one or more quantum bits. The processor **112** may determine the amount of bias and the particular quantum bits using the mapping **128** or the model **136**. The amount of bias and the quantum bits may be configured to influence the result of a computing operation **110**, such as a redirection function, of the navigation service. For example, the amount of bias and the quantum bits selected by the processor **112** may be configured to cause the navigation service to redirect the driver around the car accident using a preferred secondary route, rather than a default secondary route. In this way, the system **100** can be said to bias the computing operation **110** toward a desired result.

(31) Another example may occur within the context of an access control service. In one such example, the processor **112** may detect an event **138** (e.g., via the event detector **106**) involving an intrusion attempt from a particular network address. In response to detecting the event **138**, the processor **112** may determine an amount of bias to apply to one or more quantum bits. The amount of bias and the quantum bits may be configured to influence the result of a computing operation **110**, such as an access control function, of the access control service. For example, the amount of bias and the quantum bits selected by the processor **112** may be configured to cause the access control service to reject all login attempts from that particular network address, rather than follow a default authentication process.

(32) The system can also be used to bias the computing operation **110** away from an undesirable result. For example, in the navigation example described above, the amount of bias and the quantum bits selected by the processor **112** (e.g., using the mapping **128** or the model **136**) may be configured to cause the navigation service to direct the driver along any route that excludes the car accident. In this way, the system **100** can be said to bias the computing operation **110** away from routes involving the car accident, rather than toward any specific route. As another example, the software service **108** may be a virus scanner usable for cybersecurity purposes. In one such example, the amount of bias and the quantum bits may be configured to cause the virus scanner to prevent a user from accessing an infected file. In this way, the system **100** can be said to bias the computing operation **110** away from the infected file, rather than toward any specific file.

(33) Although FIG. **1** shows a certain number and combination of components, this is intended to be illustrative and non-limiting. Other examples may include more components, fewer components, different components, or a different arrangement of the components than is shown in FIG. **1**. For instance, although the quantum computing subsystem **116** is shown as being separate from the computer **102** in FIG. **1**, in other examples the quantum computing subsystem **116** can be part of the computer **102**.

(34) FIG. **3** is a block diagram of an example of a system **300** for implementing some aspects of the

present disclosure. The system **300** includes a processor **112** communicatively coupled to a memory **126**. The processor **112** can execute instructions **306** stored in memory **126** to perform operations. In some examples, the instructions **306** may include the event detector **106** of FIG. **1**. The instructions **306** can enable the processor **112** to perform some or all of the functionality described herein.

(35) For example, the processor **112** can detect an event **138** related to a service **108** executing in a computing environment **104**. The service **108** can be configured to rely on a state **302** of a quantum bit **120** of a quantum computing subsystem **116** in performing a computing operation **110**. This reliance is depicted in FIG. **2** via a dashed arrow. Next, the processor **112** can determine an amount of bias **304** to apply to the quantum bit **120** based on the event **138**, for example using a mapping or a model. The amount of bias **304** can be configured to modify a result **308** of the computing operation **110**. The processor **112** can then transmit a control signal **130** configured to cause the quantum computing subsystem **116** to apply the amount of bias **304** to the quantum bit **120**. The bias **304** can be applied while the service **108** is executing in the computing environment **104**.

(36) FIG. **4** is a flow chart of an example of a process for dynamically adjusting biases on quantum bits in response to detected events according to some aspects of the present disclosure. Other examples may include more operations, fewer operations different operations, or a different order of the operations than is shown in FIG. **4**. The operations of FIG. **4** are described below with reference to the components of FIGS. **1-3** above.

(37) In block **402**, the processor **112** detects an event **138** related to a software service **108** executing in a computing environment **104**. The software service **108** can be configured to rely on a state **302** of a quantum bit **120** of a quantum computing subsystem **116** in performing a computing operation **110**. The processor **112** may detect the event **138** by executing an event detector **106**.

(38) In block **404**, the processor **112** determines an amount of bias **304** to apply to the quantum bit **120** based on the event **138**. For example, the processor **112** may apply a mapping **128** or a model **136** to determine the amount of bias **304** to apply. The amount of bias **304** can be configured to modify a result **308** of the computing operation **110**.

(39) In some examples, the processor **112** also determines a quantum bit **120** to which to apply the amount of bias **304**. For example, the processor **112** may apply the mapping **128** or the model **136** to determine the quantum bit **120** to which to apply the amount of bias **304**.

(40) In block **406**, the processor **112** transmits a control signal **130** configured to cause the quantum computing subsystem **116** to apply the amount of bias **304** to the quantum bit **120**. The bias **304** can be applied while the software service **108** is executing in the computing environment **104**. The processor **112** may generate the control signal based on the amount of bias **304**, the quantum bit **120**, or both determined in block **404**.

(41) FIG. **5** is a block diagram of an example of a system **500** for controlling access to a resource by dynamically adjusting biases on quantum bits according to some aspects of the present disclosure. The system **500** includes a computer **102** that may have similar components to those described above with respect to FIG. **1**. For example, the computer **102** can include the processor **112**, memory **126**, mapping **128**, model **136**, quantum interface **114**, or any combination of these, described above. Many of those components have been intentionally omitted from FIG. **5** for simplicity.

(42) The computer **102** can be communicatively coupled to a quantum computing subsystem **116**, which may also have similar components to those described above with respect to FIG. **1**. For example, the group **118** of quantum bits can include quantum bits **120a-n** and the biasing circuit **122** can include the biasing devices **124a-n**. Many of those components have been intentionally omitted from FIG. **5** for simplicity.

(43) In this example, the system **500** is an access control system and the service is an access control service **502**. The access control service **502** can control access to one or more resources, such as a

virtual resource **510** or a physical resource **512**. Examples of the virtual resource **510** may be a secure website or a secure program. Examples of the physical resource **512** may be a secure building or a secure piece of equipment.

(44) The computer **102** can receive access control requests **508** from one or more client devices **506** or users. The access control requests **508** may be received directly or they may be received indirectly via a network, such as a local area network or the Internet. The access control requests **508** can be for obtaining access to the one or more resources. The computer **102** can provide the access control requests **508** to the access control service **502**, which can execute an access control operation **504** to adjudicate the access control requests **508**. In particular, the access control service **502** can execute the access control operation **504** to determine whether to approve or deny an access control request **508**.

(45) In some examples, the computer **102** may detect events and dynamically adjust biases on quantum bits in response to the events. These adjustments may influence results from the access control operation **504**. For example, the computer **102** may detect an event **138** using the event detector **106**. In response to detecting the event, the computer **102** can determine an amount of bias and a quantum bit (within the group **118** of quantum bits) to which to apply the amount of bias. The computer **102** may make this determination using the mapping **128**, the model **136**, or both as described above. The computer **102** may then generate and transmit one or more control signals **130** configured to cause the quantum computing subsystem **116** to apply the determined amount of bias to the determined quantum bit. More specifically, the quantum computing subsystem **116** may operate the biasing circuit **122** to apply the determined amount of bias to the determined quantum bit. This may impact how the access control operation **504** is performed and, consequently, influence a result from the access control operation **504**. Through this process, two access control requests submitted by the same client device or the same user for the same resource, at two different instants in time, may be treated in different ways by the access control service **502** depending on the events occurring between the two requests.

(46) In some aspects, biases on quantum bits can be dynamically adjusted based on detected events according to one or more of the following examples. As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”).

(47) Example #1: A non-transitory computer-readable medium of the present disclosure can include program code that is executable by a processor for causing the processor to: detect an event related to a service executing in a computing environment, the service being configured to rely on a state of a quantum bit of a quantum computing subsystem in performing a computing operation; determine an amount of bias to apply to the quantum bit based on the event, the amount of bias being configured to modify a result of the computing operation; and transmit a control signal configured to cause the quantum computing subsystem to apply the amount of bias to the quantum bit while the service is executing.

(48) Example #2: The non-transitory computer-readable medium of Example #1, wherein the service is configured to generate a first result for the computing operation prior to the amount of bias being applied to the quantum bit, and wherein the service is configured to generate a second result for the computing operation subsequent to the amount of bias being applied to the quantum bit, the second result being different from the first result.

(49) Example #3: The non-transitory computer-readable medium of any of Examples #1-2, further comprising program code that is executable by the processor for causing the processor to: access a predefined mapping that correlates a plurality of events to a plurality of bias amounts to apply to one or more quantum bits, wherein the predefined mapping includes a correlation between the event and the amount of bias to apply to the quantum bit; determine the amount of bias to apply to the quantum bit based on the correlation in the predefined mapping; and generate the control signal based on the amount of bias.

(50) Example #4: The non-transitory computer-readable medium of Example #3, wherein the predefined mapping further correlates the plurality of bias amounts to a plurality of quantum bits of the quantum computing subsystem, wherein the predefined mapping includes a relationship between the amount of bias and the quantum bit, and further comprising program code that is executable by the processor for causing the processor to: select the quantum bit, from among the plurality of quantum bits, based on the relationship in the predefined mapping; and based on selecting the quantum bit, generate the control signal to include an indicator of the quantum bit.

(51) Example #5: The non-transitory computer-readable medium of any of Examples #1-4, further comprising program code that is executable by the processor for causing the processor to: determine a plurality of bias amounts to apply to a plurality of quantum bits based on the event; and transmit one or more control signals for causing the quantum computing subsystem to apply the plurality of bias amounts to the plurality of quantum bits while the service is executing.

(52) Example #6: The non-transitory computer-readable medium of any of Examples #1-5, wherein the service is an access control system for controlling access to a resource, and the computing operation is an authorization operation for approving or denying a request from a client device to access the resource.

(53) Example #7: The non-transitory computer-readable medium of any of Examples #1-6, wherein the event is external to the service.

(54) Example #8: The non-transitory computer-readable medium of Example #7, wherein the event is a virtual event occurring in the computing environment or another computing environment.

(55) Example #9: The non-transitory computer-readable medium of Example #7, wherein the event is a physical event occurring in real space.

(56) Example #10: The non-transitory computer-readable medium of any of Examples #1-6, wherein the event is internal to the service.

(57) Example #11: A method of the present disclosure can include detecting an event related to a service executing in a computing environment, the service being configured to rely on a state of a quantum bit of a quantum computing subsystem in performing a computing operation; determining an amount of bias to apply to the quantum bit based on the event, the amount of bias being configured to modify a result of the computing operation; and transmitting a control signal configured to cause the quantum computing subsystem to apply the amount of bias to the quantum bit while the service is executing. Some or all of the method can be performed by a computer.

(58) Example #12: The method of claim of Example #11, wherein the service is configured to generate a first result for the computing operation prior to the amount of bias being applied to the quantum bit, and wherein the service is configured to generate a second result for the computing operation subsequent to the amount of bias being applied to the quantum bit, the second result being different from the first result.

(59) Example #13: The method of any of Examples #11-12, further comprising: accessing a predefined mapping that correlates a plurality of events to a plurality of bias amounts to apply to one or more quantum bits, wherein the predefined mapping includes a correlation between the event and the amount of bias to apply to the quantum bit; determining the amount of bias to apply to the quantum bit based on the correlation in the predefined mapping; and generating the control signal based on the amount of bias.

(60) Example #14: The method of Example #13, wherein the predefined mapping further correlates the plurality of bias amounts to a plurality of quantum bits of the quantum computing subsystem, wherein the predefined mapping includes a relationship between the amount of bias and the quantum bit, and further comprising: selecting the quantum bit, from among the plurality of quantum bits, based on the relationship in the predefined mapping; and based on selecting the quantum bit, generating the control signal to include an indicator of the quantum bit.

(61) Example #15: The method of any of Examples #11-14, further comprising: determining a plurality of bias amounts to apply to a plurality of quantum bits based on the event; and

transmitting one or more control signals for causing the quantum computing subsystem to apply the plurality of bias amounts to the plurality of quantum bits while the service is executing.

(62) Example #16: The method of any of Examples #11-15, wherein the service is an access control system for controlling access to a resource, and the computing operation is an authorization operation for approving or denying a request from a client device to access the resource.

(63) Example #17: The method of any of Examples #11-16, wherein the event is external to the service.

(64) Example #18: The method of Example #17, wherein the event is a virtual event occurring in the computing environment or another computing environment.

(65) Example #19: The method of Example #17, wherein the event is a physical event occurring in real space.

(66) Example #20: A system of the present disclosure can include a processor communicatively coupled to a quantum computing subsystem; and a memory including instructions that are executable by the processor for causing the processor to: detect an event related to a service executing in a computing environment, the service being configured to rely on a state of a quantum bit of the quantum computing subsystem in performing a computing operation; determine an amount of bias to apply to the quantum bit based on the event, the amount of bias being configured to modify a result of the computing operation; and transmit a control signal configured to cause the quantum computing subsystem to apply the amount of bias to the quantum bit while the service is executing.

(67) Example #21: A system of the present disclosure can include a quantum computing subsystem including: a plurality of quantum bits; and a biasing circuit configured to apply a plurality of biases to the plurality of quantum bits. The system can further comprise a processor communicatively coupled to the quantum computing subsystem; and a memory including instructions that are executable by the processor for causing the processor to: detect an event related to a service executing in a computing environment, the service being configured to rely on a state of a quantum bit of the plurality of quantum bits in performing a computing operation; determine an amount of bias to apply to the quantum bit based on the event, the amount of bias being configured to modify a result of the computing operation; and transmit a control signal configured to cause the quantum computing subsystem to apply the amount of bias to the quantum bit while the service is executing.

(68) Example #22: The system of Example #21, wherein the service is configured to generate a first result for the computing operation prior to the amount of bias being applied to the quantum bit, and wherein the service is configured to generate a second result for the computing operation subsequent to the amount of bias being applied to the quantum bit, the second result being different from the first result.

(69) Example #23: The system of any of Examples #21-22, wherein the memory further includes instructions that are executable by the processor for causing the processor to: access a predefined mapping that correlates a plurality of events to a plurality of bias amounts to apply to one or more quantum bits, wherein the predefined mapping includes a correlation between the event and the amount of bias to apply to the quantum bit; determine the amount of bias to apply to the quantum bit based on the correlation in the predefined mapping; and generate the control signal based on the amount of bias.

(70) Example #24: The system of any of Example #23, wherein the predefined mapping further correlates the plurality of bias amounts to the plurality of quantum bits of the quantum computing subsystem, wherein the predefined mapping includes a relationship between the amount of bias and the quantum bit, and wherein the memory further comprises instructions that are executable by the processor for causing the processor to: select the quantum bit, from among the plurality of quantum bits, based on the relationship in the predefined mapping; and based on selecting the quantum bit, generate the control signal to include an indicator of the quantum bit.

(71) Example #25: The system of any of Examples #21-24, wherein the memory further comprises

instructions that are executable by the processor for causing the processor to: determine a plurality of bias amounts to apply to the plurality of quantum bits based on the event; and transmit one or more control signals for causing the quantum computing subsystem to apply the plurality of bias amounts to the plurality of quantum bits while the service is executing.

(72) Example #26: The system of any of Examples #21-25, wherein the service is an access control system for controlling access to a resource, and the computing operation is an authorization operation for approving or denying a request from a client device to access the resource.

(73) Example #27: The system of any of Examples #21-26, wherein the event is external to the service.

(74) Example #28: The system of Example #27, wherein the event is a virtual event occurring in the computing environment or another computing environment.

(75) Example #29: The system of Example #27, wherein the event is a physical event occurring in real space.

(76) Example #30: The system of any of Examples #21-26, wherein the event is internal to the service.

(77) Example #31: An access control system of the present disclosure can include a processor communicatively coupled to a quantum computing subsystem; and a memory including instructions that are executable by the processor for causing the processor to: detect an event relating to an access control service executing in a computing environment, the access control service being configured to rely on a state of a quantum bit of the quantum computing subsystem in performing an access control operation; determine an amount of bias to apply to the quantum bit based on the event, the amount of bias being configured to modify a result of the access control operation; and transmit a control signal configured to cause the quantum computing subsystem to apply the amount of bias to the quantum bit while the access control service is executing.

(78) Example #32: The access control system of Example #31, wherein the memory further includes instructions that are executable by the processor for causing the processor to, prior to transmitting the control signal: receive a first access control request from a user, the first access control request being for accessing a resource; and provide the first access control request as input to the access control service, the access control service being configured to perform the access control operation based on a first state of the quantum bit and thereby generate a first response to the first access control request. The instructions are further executable by the processor for causing the processor to, subsequent to transmitting the control signal: receive a second access control request from the user, the second access control request being for accessing the resource; and provide the second access control request as input to the access control service, the access control service being configured to perform the access control operation based on a second state of the quantum bit and thereby generate a second response to the second access control request, the second state being different from the first state, and the second response being different from the first response.

(79) Example #33: The access control system of Example #32, wherein the resource is a computing resource associated with the computing environment.

(80) Example #34: The access control system of Example #32, wherein the resource is a physical resource that is external to the computing environment and located in real space.

(81) Example #35: A system of the present disclosure can include means for detecting an event related to a service executing in a computing environment, the service being configured to rely on a state of a quantum bit of a quantum computing subsystem in performing a computing operation; means for determining an amount of bias to apply to the quantum bit based on the event; and means for transmitting a control signal configured to cause the amount of bias to be applied to the quantum bit while the service is executing for modifying a result of the computing operation.

(82) The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or

to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure. For instance, any example described herein can be combined with any other examples to yield further examples.

Claims

1. A non-transitory computer-readable medium comprising program code that is executable by a processor for causing the processor to: detect an event related to a service executing in a computing environment, the service being configured to rely on a state of a quantum bit of a quantum computing subsystem in performing a computing operation, wherein the event is a physical event occurring in real space or the event is internal to the service; determine an amount of bias to apply to the quantum bit based on the event, the amount of bias being configured to modify a result of the computing operation; and transmit a control signal configured to cause the quantum computing subsystem to apply the amount of bias to the quantum bit while the service is executing.
2. The non-transitory computer-readable medium of claim 1, wherein the service is configured to generate a first result for the computing operation prior to the amount of bias being applied to the quantum bit, and wherein the service is configured to generate a second result for the computing operation subsequent to the amount of bias being applied to the quantum bit, the second result being different from the first result.
3. The non-transitory computer-readable medium of claim 1, further comprising program code that is executable by the processor for causing the processor to: access a predefined mapping that correlates a plurality of events to a plurality of bias amounts to apply to one or more quantum bits, wherein the predefined mapping includes a correlation between the event and the amount of bias to apply to the quantum bit; determine the amount of bias to apply to the quantum bit based on the correlation in the predefined mapping; and generate the control signal based on the amount of bias.
4. The non-transitory computer-readable medium of claim 3, wherein the predefined mapping further correlates the plurality of bias amounts to a plurality of quantum bits of the quantum computing subsystem, wherein the predefined mapping includes a relationship between the amount of bias and the quantum bit, and further comprising program code that is executable by the processor for causing the processor to: select the quantum bit, from among the plurality of quantum bits, based on the relationship in the predefined mapping; and based on selecting the quantum bit, generate the control signal to include an indicator of the quantum bit.
5. The non-transitory computer-readable medium of claim 1, further comprising program code that is executable by the processor for causing the processor to: determine a plurality of bias amounts to apply to a plurality of quantum bits based on the event; and transmit one or more control signals for causing the quantum computing subsystem to apply the plurality of bias amounts to the plurality of quantum bits while the service is executing.
6. The non-transitory computer-readable medium of claim 1, wherein the service is an access control system for controlling access to a resource, and the computing operation is an authorization operation for approving or denying a request from a client device to access the resource.
7. A method comprising: detecting, by a computer, an event related to a service executing in a computing environment, the service being configured to rely on a state of a quantum bit of a quantum computing subsystem in performing a computing operation; determining, by the computer, an amount of bias to apply to the quantum bit based on the event, the amount of bias being configured to modify a result of the computing operation, wherein the amount of bias is determined by: accessing a predefined mapping that correlates a plurality of events to a plurality of bias amounts to apply to one or more quantum bits, wherein the predefined mapping includes a correlation between the event and the amount of bias to apply to the quantum bit; and determining the amount of bias to apply to the quantum bit based on the correlation in the predefined mapping;

generating, by the computer, a control signal based on the amount of bias; and transmitting, by the computer, the control signal to the quantum computing system, wherein the quantum computing subsystem responds to the control signal by applying the amount of bias to the quantum bit while the service is executing.

8. The method of claim 7, wherein the service is configured to generate a first result for the computing operation prior to the amount of bias being applied to the quantum bit, and wherein the service is configured to generate a second result for the computing operation subsequent to the amount of bias being applied to the quantum bit, the second result being different from the first result.

9. The method of claim 7, wherein the predefined mapping further correlates the plurality of bias amounts to a plurality of quantum bits of the quantum computing subsystem, wherein the predefined mapping includes a relationship between the amount of bias and the quantum bit, and further comprising: selecting the quantum bit, from among the plurality of quantum bits, based on the relationship in the predefined mapping; and based on selecting the quantum bit, generating the control signal to include an indicator of the quantum bit.

10. The method of claim 7, further comprising: determining a set of bias amounts to apply to a set of quantum bits based on the event; and transmitting one or more control signals for causing the quantum computing subsystem to apply the set of bias amounts to the set of quantum bits while the service is executing.

11. The method of claim 7, wherein the service is an access control system for controlling access to a resource, and the computing operation is an authorization operation for approving or denying a request from a client device to access the resource.

12. The method of claim 7, wherein the event is external to the service.

13. The method of claim 12, wherein the event is a virtual event occurring in the computing environment or another computing environment.

14. The method of claim 12, wherein the event is a physical event occurring in real space.

15. A system comprising: a processor communicatively coupled to a quantum computing subsystem; and a memory including instructions that are executable by the processor for causing the processor to: detect an event related to a service executing in a computing environment, the service being configured to rely on a state of a quantum bit of the quantum computing subsystem in performing a computing operation, wherein the service is an access control service for controlling access to a resource, and wherein the computing operation is an authorization operation for approving or denying a request from a client device to access the resource; determine an amount of bias to apply to the quantum bit based on the event, the amount of bias being configured to modify a result of the computing operation; and transmit a control signal configured to cause the quantum computing subsystem to apply the amount of bias to the quantum bit while the service is executing.

16. The system of claim 15, wherein the service is configured to generate a first result for the computing operation prior to the amount of bias being applied to the quantum bit, and wherein the service is configured to generate a second result for the computing operation subsequent to the amount of bias being applied to the quantum bit, the second result being different from the first result.

17. The system of claim 15, wherein the memory further includes instructions that are executable by the processor for causing the processor to: access a predefined mapping that correlates a plurality of events to a plurality of bias amounts to apply to one or more quantum bits, wherein the predefined mapping includes a correlation between the event and the amount of bias to apply to the quantum bit; determine the amount of bias to apply to the quantum bit based on the correlation in the predefined mapping; and generate the control signal based on the amount of bias.

18. The system of claim 15, wherein the event is a physical event occurring in real space.

19. The system of claim 15, wherein the event is internal to the service.

20. The system of claim 15, wherein the memory further includes instructions that are executable

by the processor for causing the processor to: determine a plurality of bias amounts to apply to a plurality of quantum bits based on the event; and transmit one or more control signals for causing the quantum computing subsystem to apply the plurality of bias amounts to the plurality of quantum bits while the service is executing.
